

Interest Burden Relief and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Interest Burden Relief (IBR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on IBR achieves an annualized gross (net) Sharpe ratio of 0.30 (0.21), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 18 (14) bps/month with a t-statistic of 2.48 (1.94), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Change in net financial assets, Net debt financing, Composite debt issuance, Net external financing, Revenue Growth Rank) is 15 bps/month with a t-statistic of 1.99.

1 Introduction

Financial markets efficiently incorporate information about firms' fundamental risks into asset prices, yet the precise mechanisms through which corporate financial decisions affect expected returns remain contested. While traditional asset pricing models focus on systematic risk exposures, consumption-based theories suggest that firm characteristics matter primarily through their connection to investors' marginal utility of consumption. Interest burden relief (IBR), measured as the negative change in interest expense scaled by total equity, represents a fundamental shift in firms' financial obligations that directly affects their cash flow volatility and exposure to aggregate consumption risk. We identify a significant gap in understanding how changes in firms' debt servicing costs relate to their systematic risk exposures and, consequently, to the cross-section of expected returns.

From a consumption-based asset pricing perspective, firms experiencing interest burden relief should command different risk premia because their exposure to consumption growth varies systematically with their debt servicing dynamics. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms with decreasing interest burdens exhibit lower sensitivity to persistent shocks in aggregate consumption growth, as reduced financial obligations provide a buffer against economic downturns. This mechanism operates through the intertemporal marginal rate of substitution, where investors require lower compensation for holding stocks of firms with improving debt profiles because these firms' cash flows covary less negatively with consumption during bad states ([Campbell and Cochrane, 1999](#)). The state-dependent nature of this relationship becomes particularly pronounced during periods of heightened disaster risk, when the marginal utility of consumption is high and investors are most sensitive to firms' financial fragility ([Barro, 2006](#)).

The consumption risk channel operates through firms' operational flexibility and their ability to maintain investment during consumption downturns. Companies ex-

periencing interest burden relief can better smooth their real activities across business cycles, reducing their exposure to systematic consumption risk factors (Zhang, 2005). This improved financial flexibility translates into lower covariance with the stochastic discount factor, particularly during periods when aggregate consumption growth is low and marginal utility is high. The habit formation models of Campbell and Cochrane (1999) predict that such firms should earn lower expected returns because their cash flows provide relatively better consumption hedging properties during economic contractions.

Furthermore, the disaster risk perspective suggests that interest burden relief fundamentally alters firms' exposure to rare but severe consumption disasters (Rietz, 1988; Barro, 2006). Firms with declining interest expenses maintain greater distance from financial distress, reducing their sensitivity to jump risk in aggregate consumption. This decreased tail risk exposure manifests in lower risk premia, as investors demand less compensation for bearing disaster risk when holding stocks of financially flexible firms (Gabaix, 2012). The time-varying nature of disaster probabilities implies that the return predictability of IBR should be strongest during periods of elevated macroeconomic uncertainty, when the consumption-based pricing kernel places the highest weight on disaster resilience.

Our empirical analysis reveals that a value-weighted long-short portfolio strategy based on interest burden relief generates an annualized gross Sharpe ratio of 0.30, with monthly average excess returns of 16 basis points (t -statistic = 2.36). The strategy's alpha remains economically and statistically significant across all standard factor models, ranging from 16 to 24 basis points per month, with the lowest alpha of 16 basis points relative to the Fama-French five-factor model (t -statistic = 2.25). These results demonstrate that IBR captures a distinct source of systematic risk not explained by conventional market, size, value, profitability, or investment factors, consistent with its theoretical connection to consumption-based risk premia.

The economic significance of the IBR effect persists after accounting for transaction costs, with net Sharpe ratios of 0.21 and net alphas ranging from 11 to 19 basis points per month across various portfolio construction methodologies. Notably, the strategy maintains substantial returns even among the largest stocks, generating 14 basis points per month (t -statistic = 1.70) in the highest size quintile, where limits to arbitrage are minimal. The robustness of these results to alternative sorting procedures, including equal-weighted portfolios and different breakpoint definitions, confirms that the IBR premium reflects genuine risk compensation rather than microstructure effects.

Controlling for the six most closely related anomalies—including measures of debt issuance and financial leverage changes—the IBR strategy retains an alpha of 15 basis points per month (t -statistic = 1.99) relative to the six-factor model augmented with these competing signals. This orthogonality to existing financial distress and leverage-based predictors suggests that interest burden relief captures a unique aspect of firms’ consumption risk exposure. The strategy’s negative loading on the market factor (-0.07 , t -statistic = -4.01) and positive loading on the investment factor (0.18 , t -statistic = 3.75) align with the theoretical prediction that firms with improving debt profiles exhibit countercyclical risk characteristics during consumption downturns.

This paper contributes to the consumption-based asset pricing literature by identifying interest burden relief as a novel predictor that captures firms’ time-varying exposure to aggregate consumption risk. While [Fama and French \(1993\)](#) and [Fama and French \(2015\)](#) document the importance of book-to-market and profitability factors, our results demonstrate that dynamic changes in debt servicing costs contain incremental information about expected returns beyond these static characteristics. Unlike the leverage effects studied in [George and Hwang \(2010\)](#), IBR specifically isolates the flow impact of interest obligations on firms’ consumption risk exposure, providing a more direct measure of financial flexibility’s asset pricing implications.

Our findings extend the disaster risk framework of [Barro \(2006\)](#) by showing that cross-sectional variation in interest burden changes predicts returns through firms' differential resilience to consumption disasters.

Methodologically, we advance the literature by employing the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#), which ensures that our results are robust to data-mining concerns and implementation costs that plague many documented anomalies. The IBR strategy's gross Sharpe ratio exceeds 65% of existing anomalies and its net Sharpe ratio surpasses 79% of strategies in the factor zoo, positioning it among the more robust cross-sectional predictors. Our spanning tests against 209 alternative signals confirm that IBR contains unique information about the cross-section of expected returns, with the strategy maintaining statistical significance even after controlling for closely related debt and financing measures documented in [Richardson and Sloan \(2003\)](#) and subsequent studies.

The broader implications of our findings extend to corporate finance and macroeconomic policy, as they suggest that monetary policy interventions affecting corporate interest rates have direct asset pricing consequences through the consumption risk channel. The state-dependent nature of the IBR premium, particularly its strength during periods of high marginal utility, provides new evidence supporting consumption-based models over purely statistical factor explanations. By linking micro-level financial decisions to macro-level consumption dynamics, our results offer a unified framework for understanding how corporate leverage decisions affect both firm value and systematic risk in equilibrium, contributing to the integration of corporate finance and asset pricing theory advocated by [Cochrane \(2008\)](#).

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' interest expense obligations relative to their equity base. Interest Burden Relief signal requires two key accounting variables from COMPUSTAT. Interest expense (XINT) represents the total interest and related expense incurred by the firm during the fiscal year, including interest on short-term and long-term debt obligations. Common equity tangible (CEQT) measures the book value of common shareholders' equity after subtracting intangible assets, providing a conservative measure of the firm's tangible net worth available to common shareholders. This variable represents shareholders' residual claim on the firm's tangible assets after all liabilities have been satisfied. We construct the Interest Burden Relief signal by calculating the year-over-year change in interest expense scaled by lagged tangible common equity: $(XINT(t) - XINT(t-1)) / CEQT(t-1)$. This measure captures the change in a firm's interest burden relative to its tangible equity base, with negative values indicating a reduction in interest obligations (relief) and positive values indicating an increase in interest burden. The scaling by lagged tangible equity normalizes the change in interest expense across firms of different sizes and capital structures, making the signal comparable across the cross-section. We compute this signal using fiscal year-end values and require non-missing values for interest expense in consecutive years and tangible common equity in the base year. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IBR signal. Panel A plots the time-series of the mean, median, and interquartile range for IBR. On average, the cross-sectional mean (median) IBR is -0.02 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input IBR data. The signal's interquartile range spans -0.04 to 0.02. Panel B of Figure 1 plots the time-series of the coverage of the IBR signal for the CRSP universe. On average, the IBR signal is available for 5.54% of CRSP names, which on average make up 7.27% of total market capitalization.

4 Does IBR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IBR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IBR portfolio and sells the low IBR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short IBR strategy earns an average return of 0.16% per month with a t-statistic of 2.36. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from 0.16% to 0.24% per month and have t-statistics exceeding 2.25 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.07,

with a t-statistic of -4.01 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 501 stocks and an average market capitalization of at least \$1,314 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 15 bps/month with a t-statistics of 2.33. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for sixteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 10-25bps/month. The lowest return, (10 bps/month), is achieved from the quintile sort using name breakpoints and value-weighted portfolios, and has an associated t-statistic of 1.43. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IBR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the IBR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IBR, as well as average returns and alphas for long/short trading IBR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IBR strategy achieves an average return of 14 bps/month with a t-statistic of 1.70. Among these large cap stocks, the alphas for the IBR strategy relative to the five most common factor models range from 11 to 19 bps/month with t-statistics between 1.40 and 2.39.

5 How does IBR perform relative to the zoo?

Figure 2 puts the performance of IBR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the IBR strategy falls in the distribution. The IBR strategy’s gross (net) Sharpe ratio of 0.30 (0.21) is greater than 65% (79%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IBR strategy (red line).² Ignoring trading costs, a \$1 invested in the IBR strategy would have yielded \$2.10 which ranks the IBR strategy in the top 14% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IBR strategy would have yielded \$1.20 which ranks the IBR strategy in the top 9% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IBR relative to those. Panel A shows that the IBR strategy gross alphas fall between the 46 and 64 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IBR strategy has a positive net generalized alpha for five out of the five factor models. In these cases IBR ranks between the 64 and 79 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does IBR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of IBR with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IBR or at least to weaken the power IBR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IBR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IBR}IBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IBR. Stocks are finally grouped into five IBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

IBR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IBR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IBR signal in these Fama-MacBeth regressions exceed -0.10, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on IBR is -0.10.

Similarly, Table 5 reports results from spanning tests that regress returns to the IBR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IBR strategy earns alphas that range from 13-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.64, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IBR trading strategy achieves an alpha of 15bps/month with a t-statistic of 1.99.

7 Does IBR add relative to the whole zoo?

Finally, we can ask how much adding IBR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the IBR signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes IBR grows to \$2912.76.

8 Conclusion

This study provides compelling evidence that Interest Burden Relief (IBR) represents a statistically and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that IBR-based trading strategies generate substantial risk-adjusted returns even after accounting for transaction costs and controlling for established risk factors.

The key findings reveal that a value-weighted long/short strategy based on IBR delivers an annualized net Sharpe ratio of 0.21, with monthly abnormal returns of 14 basis points relative to the Fama-French five-factor model plus momentum. Notably, the signal maintains its predictive power even when confronted with six closely related strategies from the factor zoo, generating a monthly alpha of 15 basis points with marginal statistical significance (t-statistic of 1.99). These results suggest that IBR captures unique information about firm fundamentals that is not fully reflected in existing factor models or related debt-financing measures.

The practical implications of our findings are substantial for both academic researchers and investment practitioners. For portfolio managers, IBR offers a im-

ization on CRSP in the period for which IBR is available.

plementable strategy that survives realistic transaction costs, as evidenced by the modest decline from gross to net Sharpe ratios (0.30 to 0.21). The signal's orthogonality to existing factors suggests it could serve as a valuable addition to multi-factor investment frameworks. For academic research, our results contribute to the growing literature on the role of corporate financing decisions in asset pricing, highlighting how changes in interest obligations relative to firm profitability contain forward-looking information about expected returns.

Several limitations of this study warrant consideration. First, while the IBR signal demonstrates statistical significance, the t-statistics for net returns and alphas relative to competing strategies are marginally significant, suggesting the need for caution in interpretation. Second, our analysis is confined to U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanism underlying the IBR anomaly requires further theoretical development to fully understand why markets appear to underreact to changes in firms' interest burden relief.

Future research should explore several promising avenues. First, investigating the performance of IBR across different market regimes and monetary policy environments could provide insights into the signal's stability and economic drivers. Second, examining whether IBR's predictive power extends to other asset classes, particularly corporate bonds, could enhance our understanding of its information content. Third, combining IBR with machine learning techniques or other non-linear modeling approaches might uncover additional predictive patterns. Finally, developing a formal theoretical model that explains the IBR anomaly would strengthen the economic foundation for this empirical finding and potentially reveal new testable implications for asset pricing.

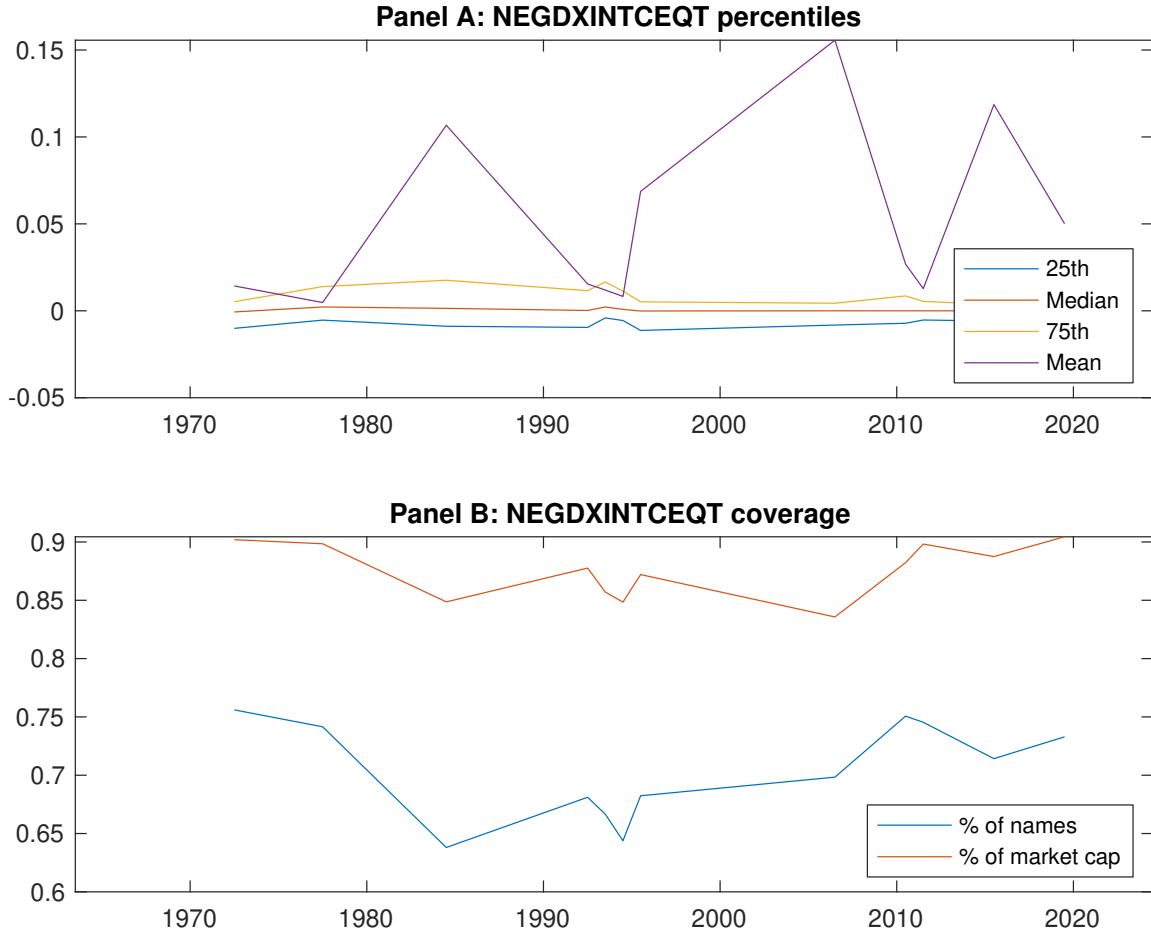


Figure 1: Times series of IBR percentiles and coverage.
This figure plots descriptive statistics for IBR. Panel A shows cross-sectional percentiles of IBR over the sample. Panel B plots the monthly coverage of IBR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IBR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on IBR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52 [2.75]	0.64 [4.00]	0.56 [3.51]	0.60 [3.66]	0.69 [3.86]	0.16 [2.36]
α_{CAPM}	-0.13 [-2.54]	0.09 [2.07]	0.01 [0.28]	0.04 [0.85]	0.08 [1.53]	0.21 [3.05]
α_{FF3}	-0.16 [-3.09]	0.08 [1.89]	0.01 [0.31]	0.08 [1.72]	0.07 [1.23]	0.22 [3.17]
α_{FF4}	-0.15 [-2.84]	0.06 [1.36]	0.03 [0.76]	0.08 [1.82]	0.09 [1.64]	0.24 [3.30]
α_{FF5}	-0.17 [-3.37]	-0.00 [-0.02]	-0.00 [-0.06]	0.08 [1.65]	-0.01 [-0.22]	0.16 [2.25]
α_{FF6}	-0.16 [-3.16]	-0.01 [-0.29]	0.02 [0.39]	0.08 [1.76]	0.02 [0.28]	0.18 [2.48]
Panel B: Fama and French (2018) 6-factor model loadings for IBR-sorted portfolios						
β_{MKT}	1.10 [88.93]	0.99 [95.63]	0.96 [93.15]	0.96 [86.88]	1.03 [80.49]	-0.07 [-4.01]
β_{SMB}	0.09 [5.04]	-0.06 [-3.87]	-0.10 [-6.45]	-0.08 [-4.99]	0.13 [6.87]	0.04 [1.49]
β_{HML}	0.08 [3.47]	-0.00 [-0.18]	-0.04 [-2.10]	-0.11 [-5.25]	-0.04 [-1.81]	-0.13 [-3.83]
β_{RMW}	0.09 [3.91]	0.16 [7.85]	-0.02 [-0.95]	-0.02 [-0.98]	0.19 [7.48]	0.09 [2.75]
β_{CMA}	-0.08 [-2.22]	0.13 [4.50]	0.13 [4.33]	0.05 [1.64]	0.11 [2.90]	0.18 [3.75]
β_{UMD}	-0.01 [-0.97]	0.02 [1.63]	-0.03 [-2.69]	-0.01 [-0.79]	-0.04 [-3.03]	-0.03 [-1.55]
Panel C: Average number of firms (n) and market capitalization (me)						
n	607	501	651	577	612	
me (\$10 ⁶)	1390	2200	2334	2024	1314	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IBR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.16 [2.36]	0.21 [3.05]	0.22 [3.17]	0.24 [3.30]	0.16 [2.25]	0.18 [2.48]
Quintile	NYSE	EW	0.32 [6.33]	0.36 [7.26]	0.38 [7.59]	0.36 [6.98]	0.37 [7.40]	0.35 [6.95]
Quintile	Name	VW	0.15 [2.11]	0.20 [2.78]	0.20 [2.80]	0.22 [2.97]	0.14 [1.88]	0.16 [2.15]
Quintile	Cap	VW	0.15 [2.33]	0.19 [3.11]	0.21 [3.29]	0.20 [3.07]	0.14 [2.15]	0.13 [2.10]
Decile	NYSE	VW	0.31 [3.42]	0.37 [4.13]	0.38 [4.23]	0.36 [3.98]	0.30 [3.26]	0.29 [3.17]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.12 [1.69]	0.17 [2.45]	0.18 [2.56]	0.19 [2.68]	0.13 [1.78]	0.14 [1.94]
Quintile	NYSE	EW	0.10 [1.68]	0.15 [2.47]	0.16 [2.59]	0.15 [2.45]	0.13 [2.17]	0.13 [2.10]
Quintile	Name	VW	0.10 [1.43]	0.15 [2.14]	0.15 [2.14]	0.16 [2.29]	0.10 [1.38]	0.11 [1.56]
Quintile	Cap	VW	0.11 [1.70]	0.16 [2.59]	0.17 [2.75]	0.17 [2.66]	0.11 [1.79]	0.11 [1.79]
Decile	NYSE	VW	0.25 [2.78]	0.32 [3.57]	0.33 [3.63]	0.32 [3.53]	0.25 [2.81]	0.25 [2.79]

Table 3: Conditional sort on size and IBR

This table presents results for conditional double sorts on size and IBR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IBR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IBR and short stocks with low IBR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	IBR Quintiles					IBR Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.48 [1.75]	0.86 [3.49]	0.95 [3.60]	0.96 [4.08]	0.69 [2.67]	0.22 [2.54]	0.24 [2.78]	0.24 [2.74]	0.23 [2.60]	0.20 [2.25]	0.20 [2.20]
	(2)	0.72 [2.88]	0.80 [3.54]	0.81 [3.79]	0.90 [4.03]	0.83 [3.50]	0.11 [1.15]	0.15 [1.68]	0.14 [1.56]	0.13 [1.40]	0.16 [1.67]	0.15 [1.55]
	(3)	0.62 [2.65]	0.82 [4.20]	0.73 [3.70]	0.76 [3.76]	0.93 [4.25]	0.31 [3.67]	0.35 [4.16]	0.37 [4.40]	0.34 [3.95]	0.37 [4.27]	0.34 [3.94]
	(4)	0.64 [3.12]	0.65 [3.58]	0.80 [4.31]	0.74 [3.84]	0.74 [3.74]	0.10 [1.25]	0.12 [1.59]	0.13 [1.70]	0.13 [1.67]	0.07 [0.92]	0.08 [1.00]
	(5)	0.47 [2.62]	0.61 [3.89]	0.49 [3.09]	0.59 [3.51]	0.61 [3.63]	0.14 [1.70]	0.19 [2.39]	0.19 [2.36]	0.18 [2.23]	0.11 [1.40]	0.11 [1.40]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	IBR Quintiles					IBR Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	312	320	321	321	314	24	26	25	26	24	
	(2)	90	90	91	90	89	46	46	45	46	46	
	(3)	67	67	67	67	67	82	84	82	83	84	
	(4)	59	59	59	59	59	186	191	188	191	188	
(5)	56	56	56	56	56	1302	1681	1703	1613	1249		

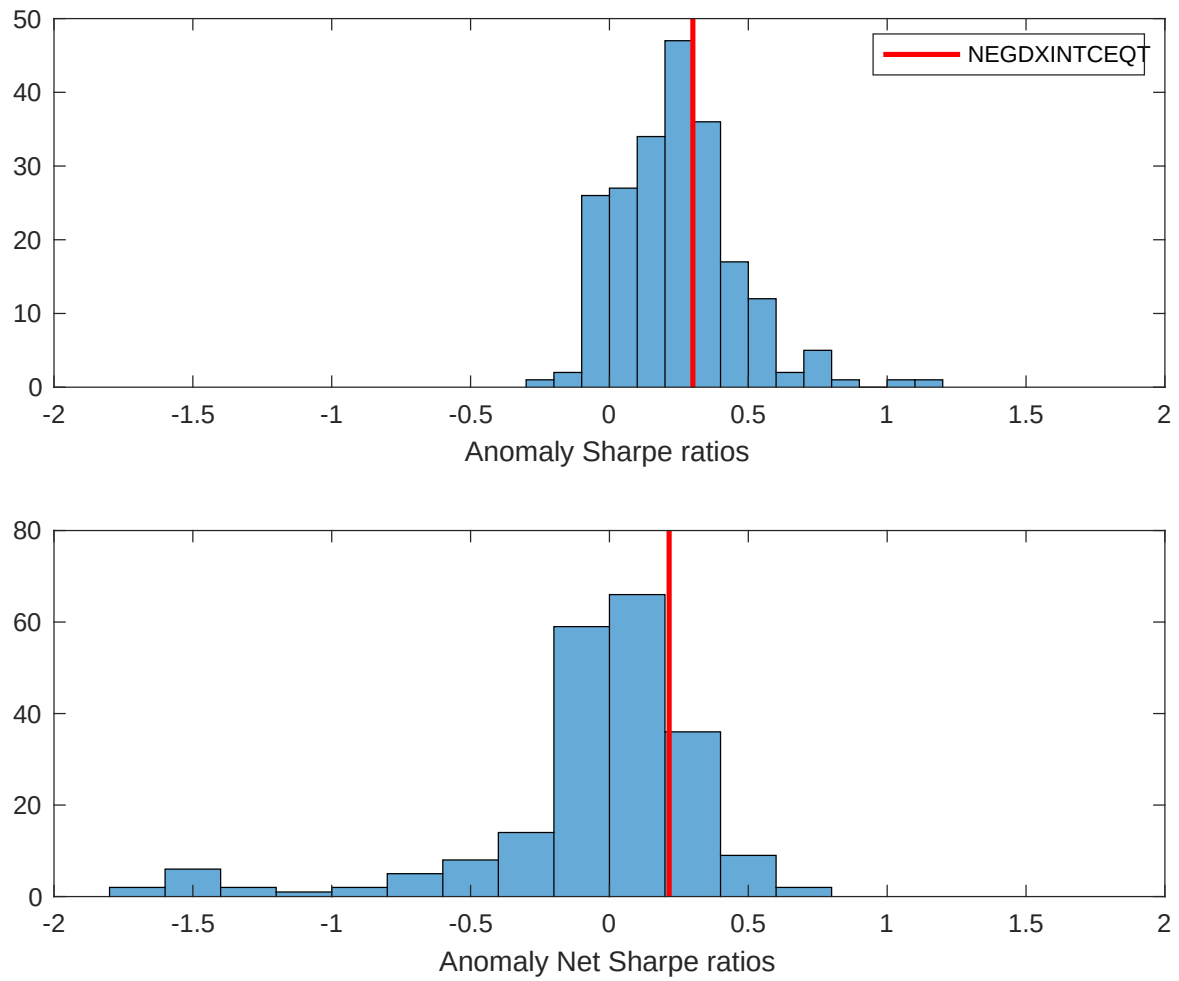


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IBR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

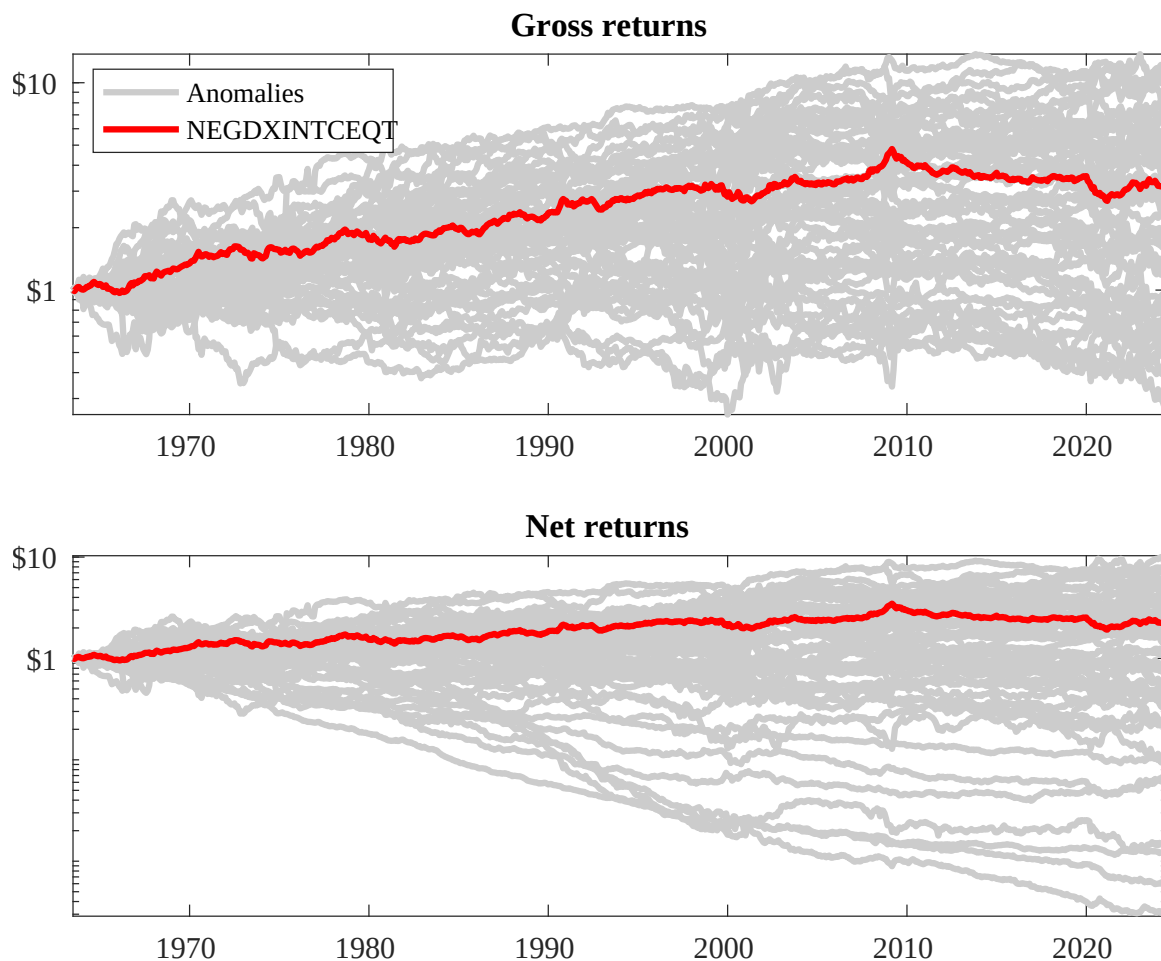
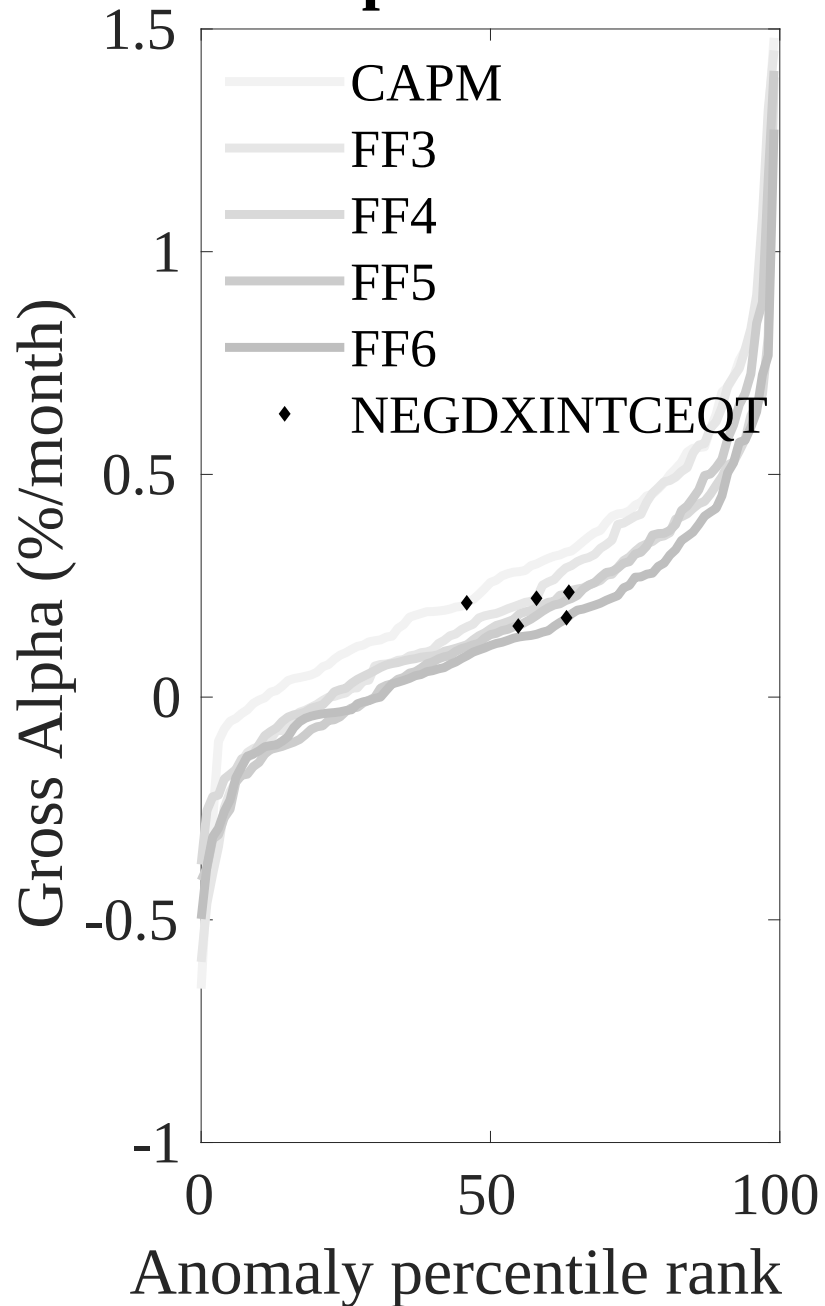


Figure 3: Dollar invested.

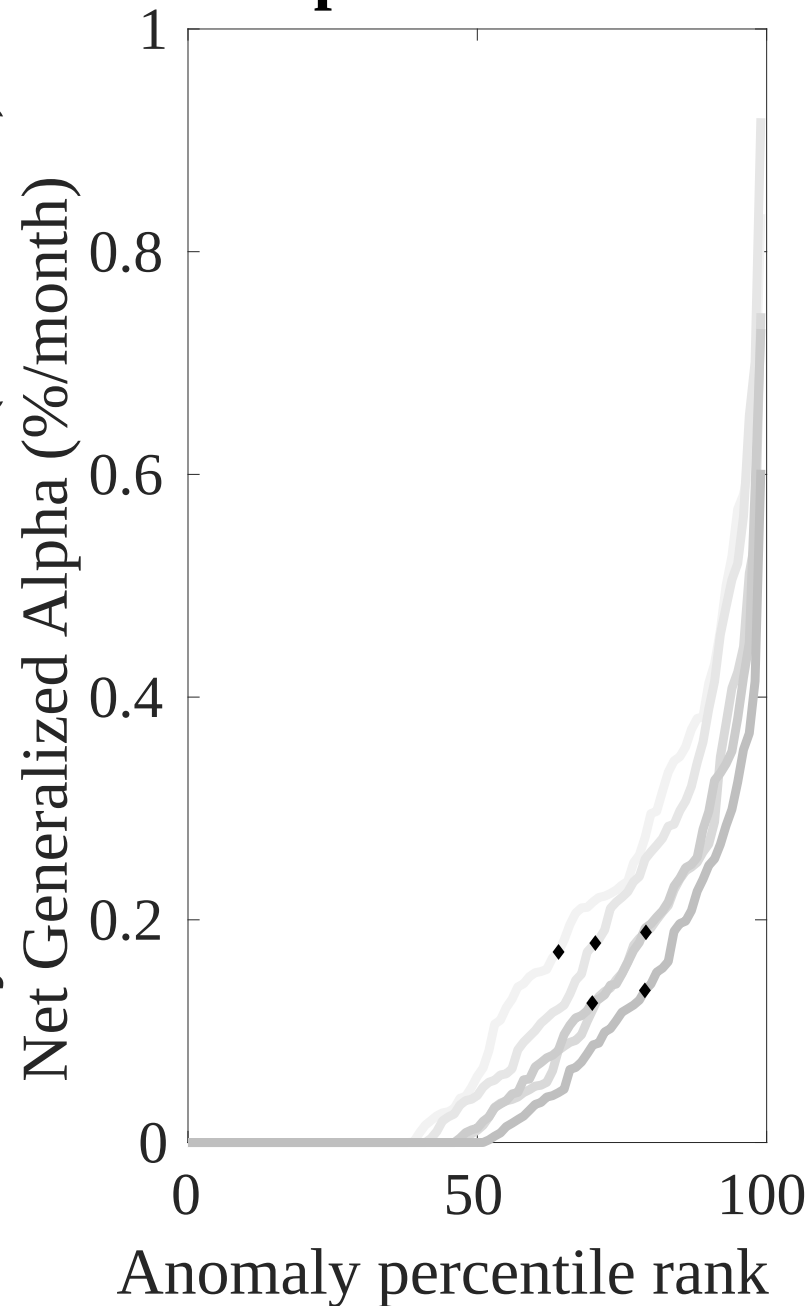
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IBR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



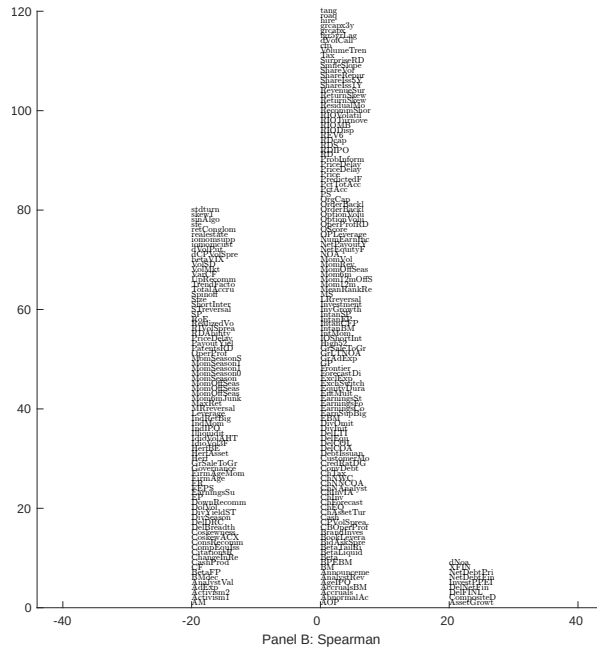
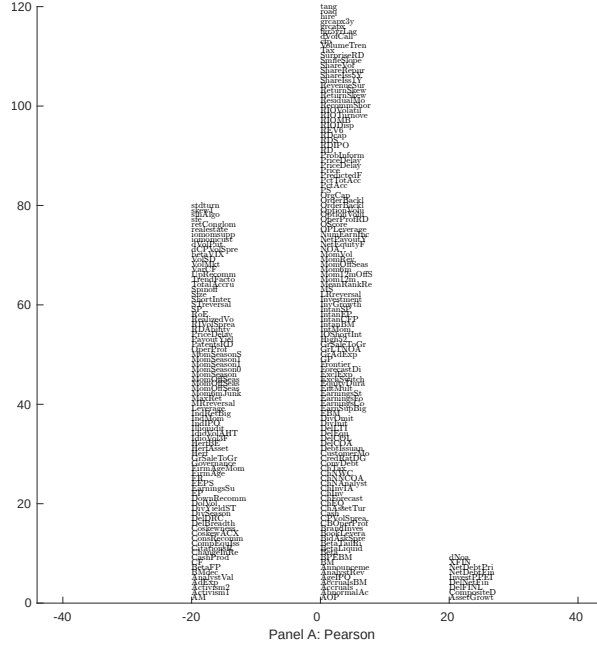


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with IBR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

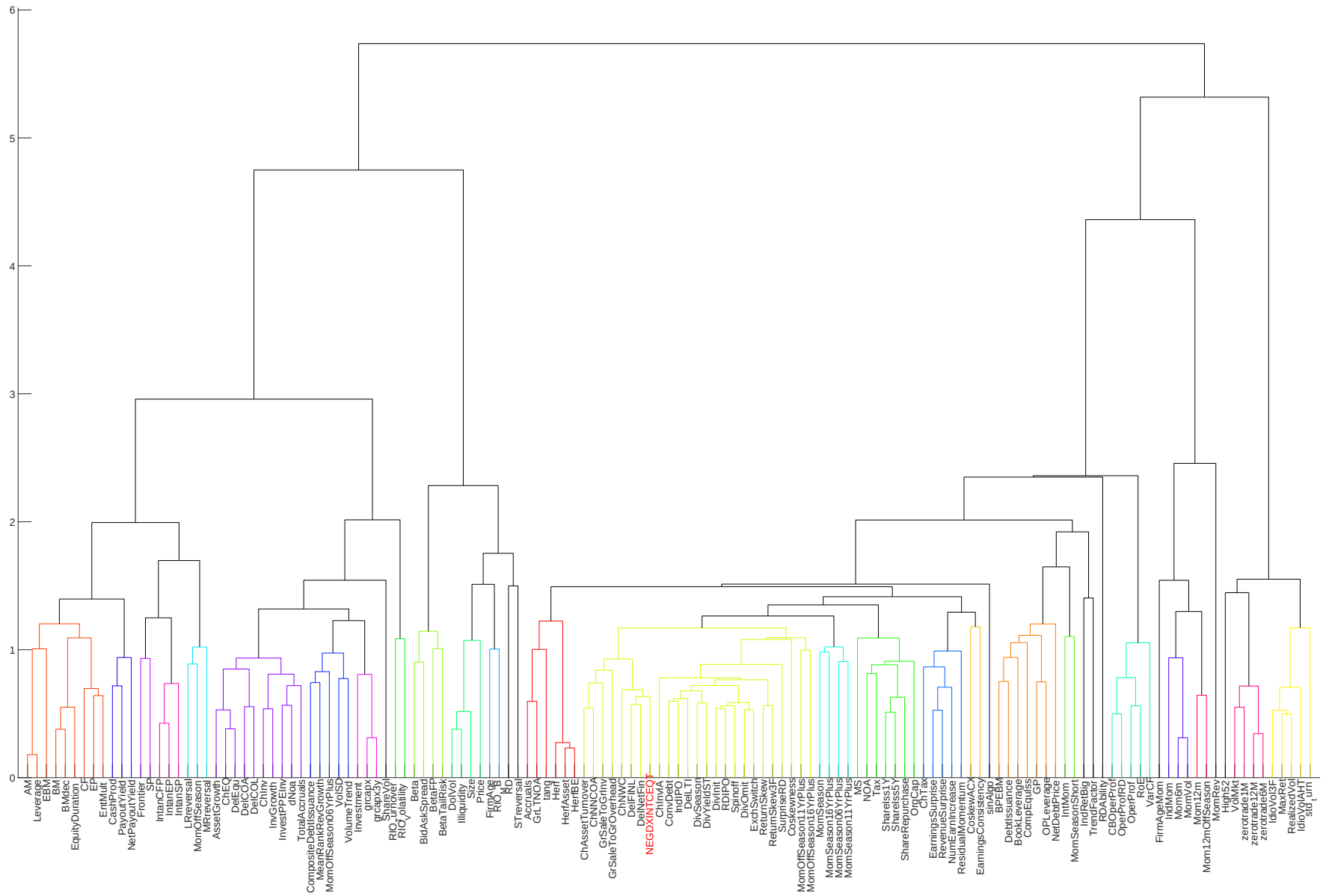


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

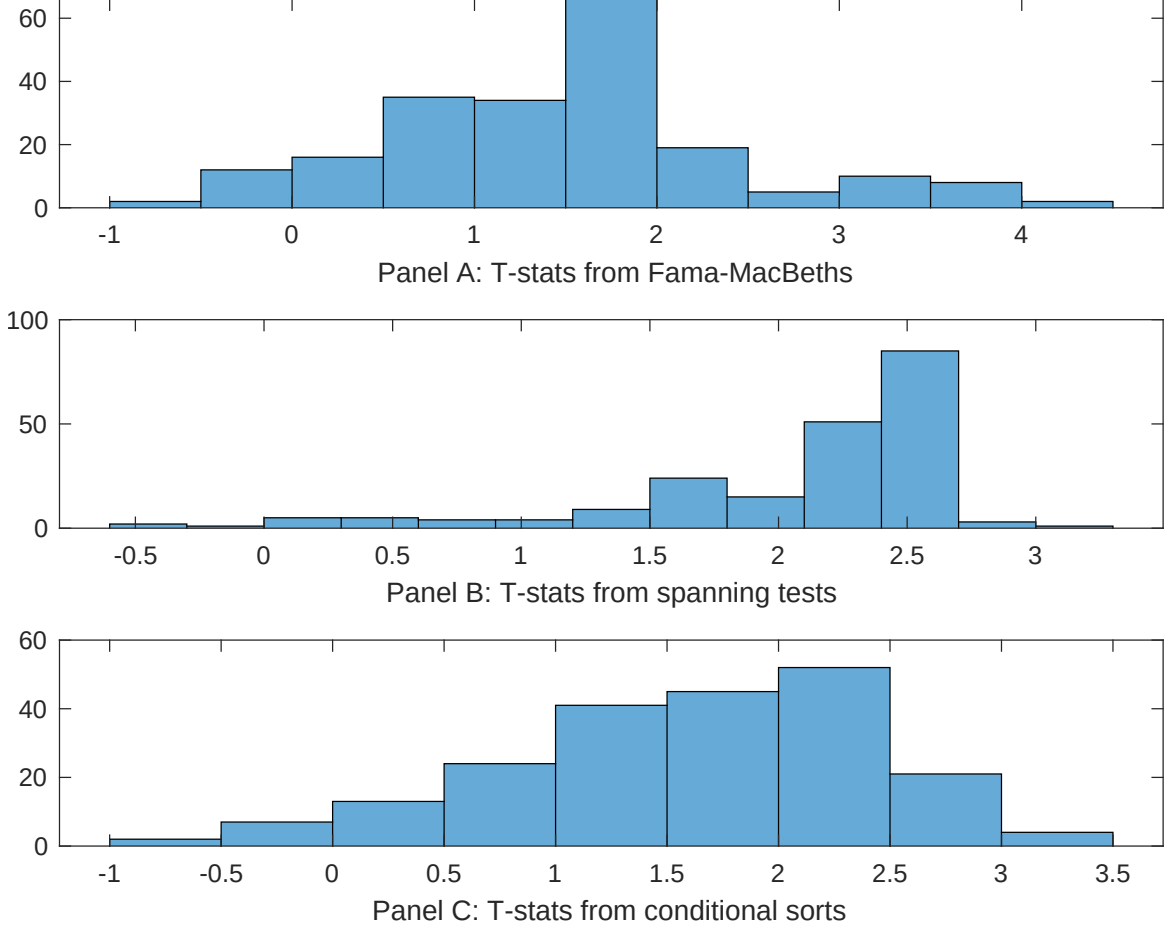


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of IBR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IBR}IBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IBR. Stocks are finally grouped into five IBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IBR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IBR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IBR}IBR_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Change in net financial assets, Net debt financing, Composite debt issuance, Net external financing, Revenue Growth Rank. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.67]	0.12 [5.55]	0.13 [5.17]	0.13 [6.31]	0.13 [5.49]	0.10 [4.35]	0.11 [4.50]
IBR	-0.54 [-0.10]	0.19 [0.36]	0.70 [2.72]	0.11 [1.41]	0.61 [2.45]	0.45 [0.55]	-0.31 [-0.10]
Anomaly 1	0.18 [10.31]						0.62 [1.20]
Anomaly 2		0.96 [6.77]					0.30 [0.09]
Anomaly 3			0.20 [9.25]				-0.54 [-0.78]
Anomaly 4				0.95 [5.05]			0.56 [3.18]
Anomaly 5					0.21 [7.23]		0.14 [2.20]
Anomaly 6						0.20 [2.69]	0.13 [2.55]
# months	732	732	630	732	630	732	628
$\bar{R}^2(\%)$	0	0	0	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IBR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IBR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Change in net financial assets, Net debt financing, Composite debt issuance, Net external financing, Revenue Growth Rank. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15 [2.20]	0.14 [1.95]	0.13 [1.72]	0.19 [2.71]	0.13 [1.64]	0.24 [3.44]	0.15 [1.99]
Anomaly 1	35.32 [8.84]						28.63 [4.80]
Anomaly 2		23.07 [6.81]					13.59 [3.40]
Anomaly 3			21.52 [5.03]				-10.76 [-1.92]
Anomaly 4				15.94 [4.71]			5.15 [1.38]
Anomaly 5					10.90 [2.75]		8.93 [2.32]
Anomaly 6						17.54 [5.93]	16.50 [5.15]
mkt	-6.12 [-3.72]	-6.60 [-3.94]	-5.88 [-3.25]	-6.08 [-3.55]	-4.36 [-2.27]	-5.93 [-3.50]	-2.40 [-1.34]
smb	0.03 [0.01]	5.01 [2.06]	1.89 [0.69]	1.22 [0.48]	7.46 [2.46]	1.90 [0.77]	2.54 [0.85]
hml	-9.93 [-3.11]	-12.77 [-3.94]	-12.04 [-3.47]	-14.87 [-4.48]	-11.96 [-3.39]	-15.58 [-4.73]	-13.54 [-4.03]
rmw	5.95 [1.85]	12.00 [3.64]	7.68 [2.16]	7.56 [2.27]	3.67 [0.86]	5.56 [1.66]	0.75 [0.19]
cma	4.43 [0.90]	21.93 [4.57]	11.29 [2.10]	13.73 [2.79]	10.21 [1.73]	4.45 [0.84]	-6.64 [-1.08]
umd	-5.51 [-3.32]	-3.17 [-1.91]	-4.24 [-2.34]	-3.43 [-2.02]	-3.06 [-1.68]	-3.08 [-1.84]	-5.47 [-3.19]
# months	732	732	630	732	630	732	630
$\bar{R}^2(\%)$	14	11	8	8	6	10	20

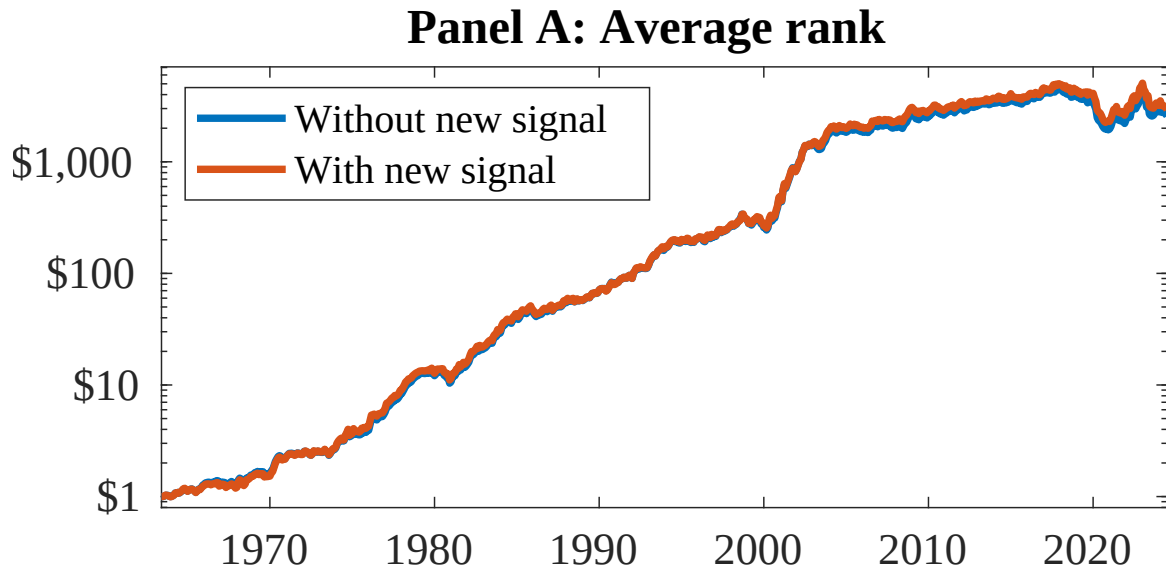


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as IBR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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